

1	(a) work done in moving unit mass from infinity (to the point)	M1 A1	[2]
	(b) (i) gravitational potential energy = GMm / x energy = $(6.67 \times 10^{-11} \times 7.35 \times 10^{22} \times 4.5) / (1.74 \times 10^6)$ energy = 1.27×10^7 J	M1 A0	[1]
	(ii) <u>change in</u> grav. potential energy = <u>change in</u> kinetic energy $\frac{1}{2} \times 4.5 \times v^2 = 1.27 \times 10^7$ $v = 2.4 \times 10^3 \text{ m s}^{-1}$	B1 A1	[2]
	(c) Earth would attract the rock / potential at Earth('s surface) not zero / < 0 / at Earth, potential due to Moon not zero escape speed would be lower	M1 A1	[2]